

Renewable Energies Pre-FEED Report

Autonomous Island Hybrid Energy System

Group **XX** – Assigned Island: **Island Name**

Student Name	Student ID	% Contribution*
Student Name 1	xxxxxx	25%
Student Name 2	xxxxxx	25%
Student Name 3	xxxxxx	25%
Student Name 4	xxxxxx	25%

*The sum of all contributions must equal 100%.

1 Island Baseline & Demand Profile

Table 1: Grid Constraints & Baseline HFO System Economics

Original Plant Capacity (MW)	X.X MW
Annual Energy Production -AEP (MWh)	XXXXXX MWh
Baseline 100% HFO LCOE (€/MWh)	XXX.XX €/MWh

Insert Monthly Demand Profile Plot Here
Monthly Energy Load (MWh) vs. Month (Jan–Dec)

Figure 1: Monthly electrical energy consumption profile across the 12 calendar months for the assigned island.

2 Renewable Resource Sourcing & Yield Modeling

2.1 A. Onshore Wind Resource

Table 2: Wind Site Parameters & Single Turbine Yield Metrics

Mean Wind Speed, $\overline{U}$ (m/s)	X.X m/s	Terrain Roughness, z_0 (m)	X.XXX m
Weibull c & k	c=X.X, k=X.X	Hub Height (m)	XXX m
Capacity Factor, C_f (%)	XX.X%	IEC Wind Class	Class I/II/III
Time % at Rated Power	XX.X%	Turbine Model	Make & Model
Time % Turbine Off	XX.X%	(Below cut-in or above cut-off)	
Min. Turbines for 100% Demand	XX turbines (To cover annual AEP on average)		

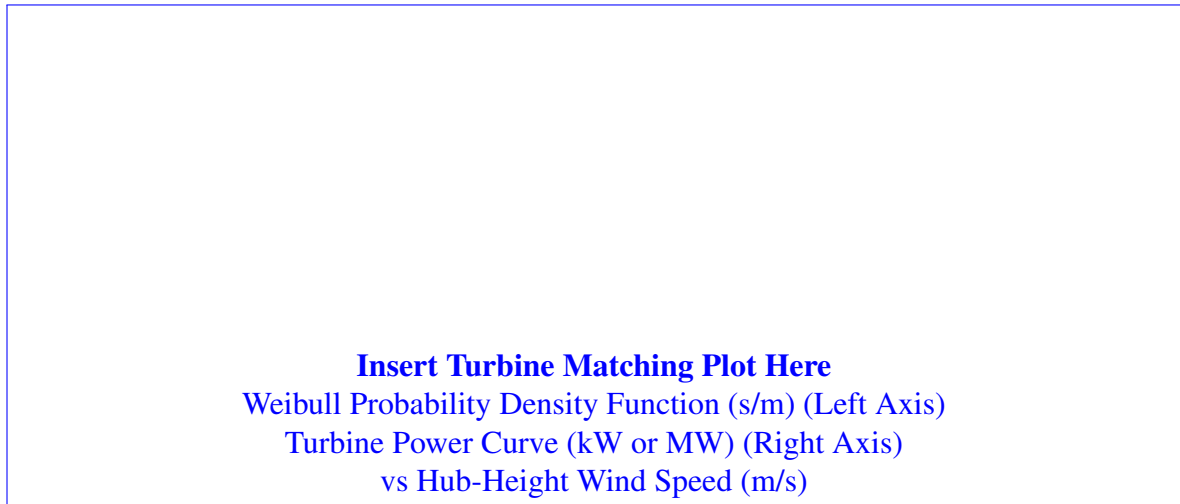


Figure 2: Weibull wind speed distribution overlaid with the selected commercial power curve.

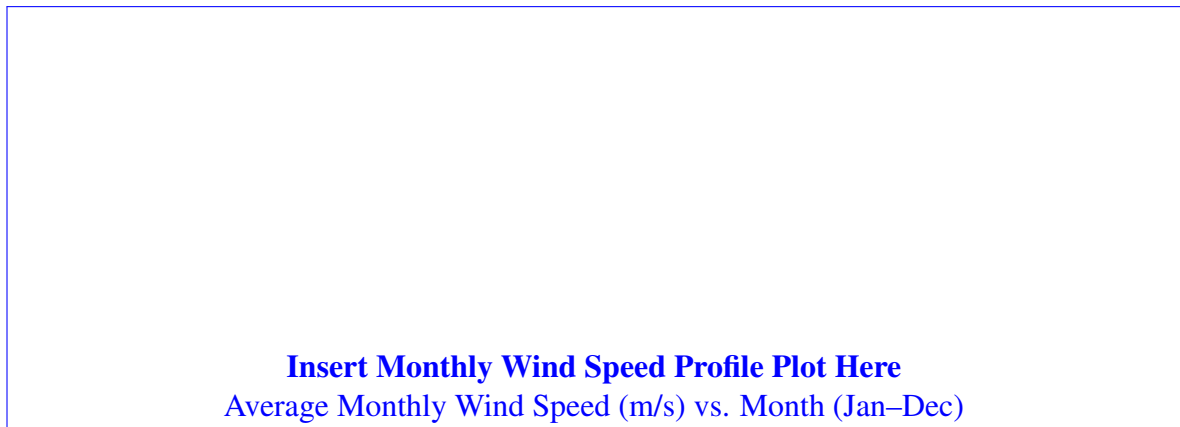


Figure 3: Monthly average wind speed at hub height across the 12 calendar months.

Table 3: Wind Resource Pre-Presentation Analysis

Observation Task (To be justified during the oral defense)	Identified Month
1) Peak Curtailment Risk: Comparing the monthly wind speed profile with Figure 1 (Monthly Demand), identify the month where the highest curtailment of wind production is expected.	Month
2) Wind-Solar Seasonal Complementarity: Comparing the monthly wind speed profile with Figure 3 (Solar Yield), identify the month where solar yield best compensates for a seasonal drop in wind speed.	Month

2.2 B. Solar PV Resource

Table 4: Solar PV Siting & Transposition

Coordinates (Lat, Lon)	XX.XX, XX.XX	Land Area Requirement	XX.X ha/MWp
Optimum Tilt, β (°)	XX°	Optimum Azimuth, ρ (°)	XX°
PV Module Type	Monocrystalline/etc.	Est. Plant PR (PR_{BOS})	XX%
Min. Area for 100% Demand	XX.X ha (To cover annual AEP on average)		



Figure 4: Monthly specific AC energy yield per unit area (MWh/ha) and average daily plane-of-array irradiance, G_{POA} (kWh/m²/day), illustrating the seasonal land-use efficiency of the proposed PV system.

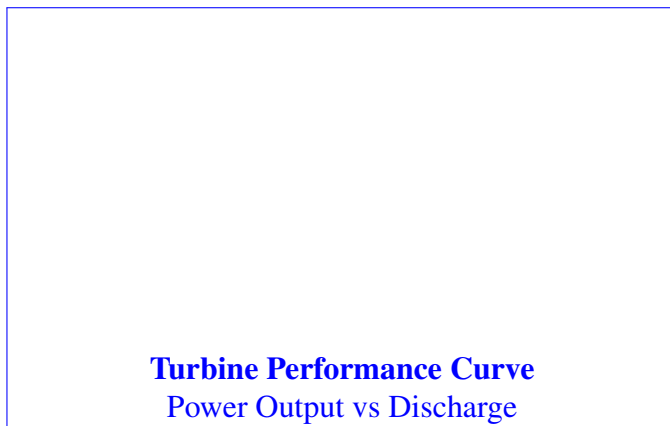
Table 5: Solar Yield Pre-Presentation Analysis

Observation Task (To be justified during the oral defense)	Identified Month
1) Maximum Deviation: Identify the month where the Monthly Energy Yield deviates the most from the G_{POA} profile.	Month
2) Peak Curtailment Risk: Comparing Figure 3 (Solar Yield) with Figure 1 (Monthly Demand), identify the month where the highest curtailment of solar production is expected.	Month

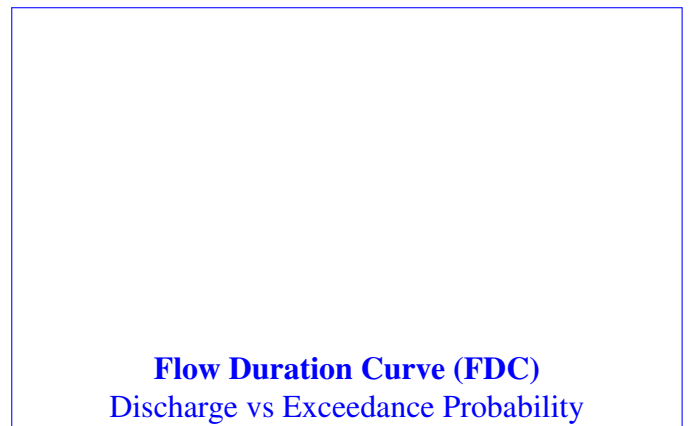
2.3 C. Run-of-River Hydropower

Table 6: Hydrological Parameters & Turbine Sizing

Gross Head (H_b)	XXX m	Design Net Head (H_n)	XXX m
Design Discharge (Q_d)	X.XX m ³ /s	FDC Exceedance Point	XX% of time
Specific Speed (N_s)	XXX	Turbine Topology	Pelton/Francis/Kaplan
Synchronous Speed (N)	XXX RPM	Generator Pole Pairs (p)	X pairs
Rated Mech. Power (P_{rated})	X.XX MW	Design Efficiency (η)	XX%



(a) Turbine Hydraulic Mapping



(b) Hydrological FDC

Figure 5: Resource mappings compressed to the 288-hour time domain.

Table 7: Hydropower Pre-Presentation Analysis

Observation Task (To be justified during the oral defense)	Identified Value
1) Wet Season Spill Percentage: Based on your chosen FDC Exceedance Point, what percentage of the year does river flow exceed your turbine's maximum capacity (resulting in spilled water)?	XX%
2) Dry Season Drought Risk: Identify the period or FDC probability range where flow is lowest. Are you relying primarily on wind, solar, or thermal backup during this hydro deficit?	Resource/Time

3 System Sizing & Dynamic Dispatch

Table 8: Proposed Hybrid Plant Installed Capacities & Penetration Metrics

Asset	Installed Capacity (MW)	Annual Energy Yield (MWh/year)
Wind Farm	X.X MW	XXXX MWh
Solar PV Plant	X.X MW	XXXX MWh
Hydro Power Plant	X.X MW	XXXX MWh
Thermal Plant (Downsized)	X.X MW	XXXX MWh (Dispatch limit)
VRE Energy Penetration (% AEP)	XX.X% (Total VRE MWh / Total Demand MWh)	
VRE Capacity Penetration (% MW)	XX.X% (Total VRE MW / Total Installed MW)	
VRE Overbuild Ratio	XXX% (Total VRE MW / Peak Demand P_{peak})	

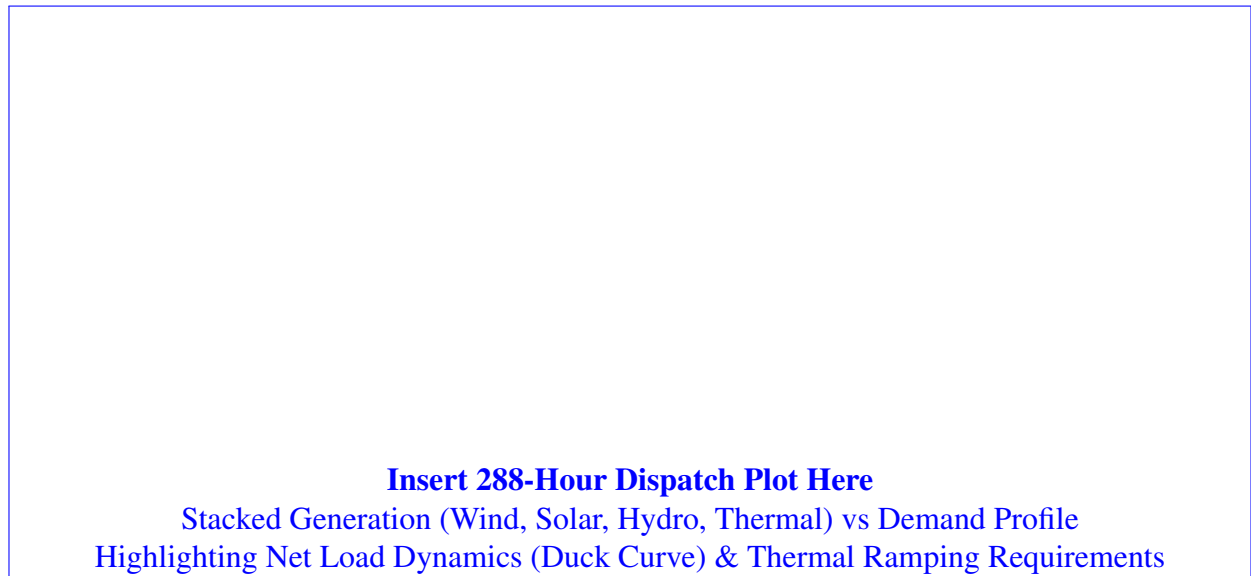


Figure 6: Hourly grid dispatch balance across the 12 representative days (288 hours) illustrating VRE contribution, thermal ramping, and net-load duck curve behavior.

Table 9: System Design & Dynamic Dispatch Analysis (Max 500 chars/row)

Topic	Discussion
Penetration & Overbuild Strategy	Compare your VRE Energy Penetration (% AEP) against Capacity Penetration (% MW). Justify the resulting Overbuild Ratio—why is overbuilding necessary for high energy yield, and what are its curtailment implications? [Max 500 chars]
Complementarity & Duck Curve	Discuss how solar, wind, and hydro profiles complement each other. Highlight any observed "duck curve" behavior and the resulting thermal ramping rates required when solar generation drops off. [Max 500 chars]
Thermal Sizing & Firm Backup	Justify the downsized capacity of your thermal peaker plant. How did you guarantee 100% grid reliability (zero unserved energy) during multi-day VRE droughts? [Max 500 chars]

4 Economic Analysis & Grid Resilience

4.1 Financial Inputs & LCOE Results

Table 10: Baseline Economics & LCOE Comparison

Project Lifespan (t)	20 Years	Discount Rate (r)	X.X%
HFO Fuel Delivered Cost	X.XXX €/L	Carbon Tax	XX.X €/tCO ₂
Baseline 100% HFO LCOE	XXX.XX €/MWh		
Scenario A LCOE (Proposed Hybrid)	XXX.XX €/MWh		
Scenario B LCOE (100% Firm Backup)	XXX.XX €/MWh		
Reliability Premium (ΔLCOE_{AB})	XX.XX €/MWh		

4.2 Sensitivity Matrix

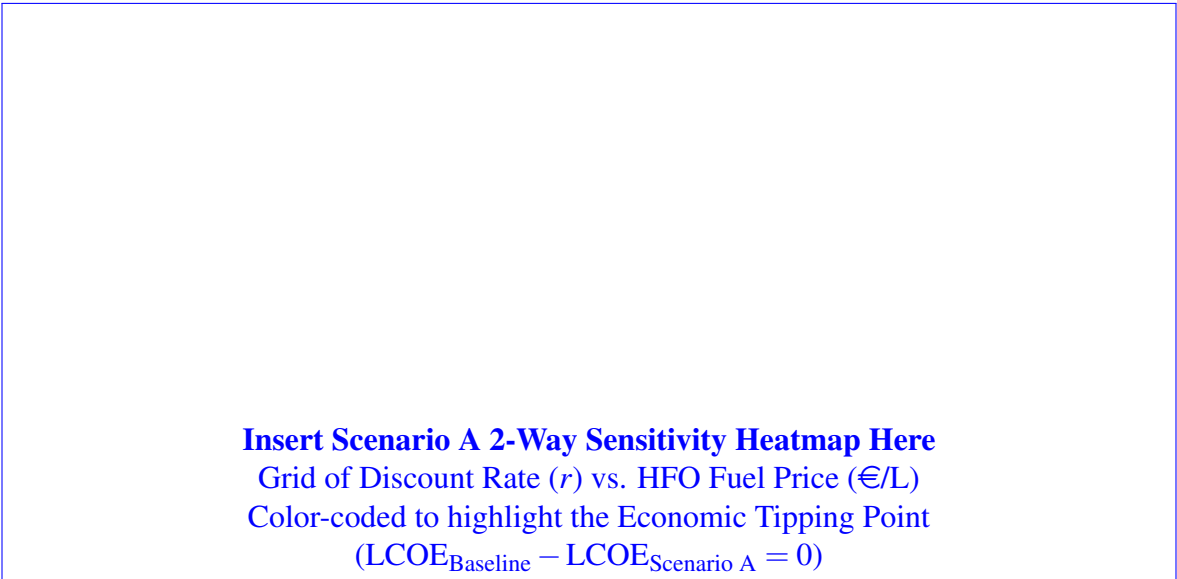


Figure 7: Color-mapped LCOE sensitivity matrix demonstrating under which financial conditions Scenario A (Proposed Hybrid) outcompetes the 100% HFO baseline.

4.3 4.3 Final Executive Summary

Table 11: Economic & Resilience Discussion (Max 500 chars/row)

Topic	Discussion
Economic Tipping Point & Fuel Price	Analyze the 2-way sensitivity heatmap. At what fuel price and discount rate (r) does Scenario A reach economic parity ($\Delta\text{LCOE} = 0$) against the baseline? How does the added Carbon Tax ($\text{€}/\text{tCO}_2$) penalty influence this feasibility threshold? [Max 500 chars]
Thermal Asset Strategy & CAPEX	State whether your downsized thermal plant in Scenario A retains legacy units (sunk asset, $\text{€}0$ CAPEX) or installs new fast-ramping peakers (new CAPEX penalty). Justify the economic impact on your LCOE. [Max 500 chars]
Reliability Premium & Recommendation	Discuss the ΔLCOE_{AB} cost premium of Scenario B ($N - 1$ full thermal reserve). Provide a final recommendation to the Island Energy Council: should they fund full backup or accept Scenario A's dynamic dispatch? [Max 500 chars]